VISHESH VISHWAKARMA

Fourth Year Undergraduate
Department of Chemical Engineering
Indian Institute of Technology Kanpur

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Examination	Degree/Certificate	Institute	Year	CPI/%
Graduation	B.Tech	Indian Institute of Technology Kanpur	2023–2027	**7.1**/10
Intermediate/+2	CBSE(XII)	Shiv Jyoti Convent School	2022	**87**%
Matriculation	CBSE(X)	Maharishi Arvind Public School	2020	**94**%

SCHOLASTIC ACHIEVEMENTS

- **WorldQuant BRAIN Grandmaster**, Global Rank 30, Q4 2024
- **MAPC Star**, Rank 36, WorldQuant BRAIN
- **Alphagrep**, Global Rank 24, Alphaverse Premier League

WORK EXPERIENCE

Quant Singularity *Jun'26 – Aug'26*
Machine Learning Intern

- Built components of a modular **quantitative backtesting and research framework**, integrating strategy abstractions, validation, experiment tracking and reproducible research workflows
- Implemented a **feature-engineering and Vault layer** with schema validation, feature lineage, metadata stamping and Parquet persistence for systematic ML research
- Developed **MLflow research infrastructure** and automated tests for experiment reproducibility and strategy evaluation

BeyondIRR *May'25 – Jul'25*
Quantitative Research Intern Mumbai

- Developed an **NSE 500 strategy with 21% CAGR (2006–2025)**, outperforming the benchmark with lower volatility using factor-based signals with alternative, technical and fundamental data
- Constructed **risk-constrained optimized portfolios**, improving long-term returns by 5% over the backtesting horizon; collaborated with research and trading teams on scenario simulation
- Improved data-pipeline latency through **database-fetch caching** and contributed to strategy evaluation and deployment pipelines

Q-DITS *Jan'25 – Apr'25*
Quant Developer Intern India

- Built **hybrid trading strategies** using technical and fundamental data across multiple stock universes and created high-frequency models on 5-minute intervals
- Applied **econometric modelling** to improve capital efficiency and optimize risk-adjusted returns by 9.2%
- Validated strategy robustness across market universes using **MATLAB simulations** and systematic backtesting

WorldQuant BRAIN *Jun'24 – Present*
Quantitative Research Consultant India

- Contributed **1100+ alpha signals** across global markets with **Sharpe > 3** and **fitness > 2** consistently beating benchmarks; created 100+ multi-factor "Super Alphas"
- Built **Python research pipelines** integrated with BRAIN APIs and genetic programming to enhance alpha signal discovery
- Applied **machine learning** for feature clustering and data cleaning, alongside rigorous robustness testing of factor-based signals
- Researched systematic opportunities across technical, fundamental, sentiment and alternative datasets for **cross-sectional alpha generation**

TECHNICAL SKILLS

- **Languages:** Python, C++, C, MATLAB, SQL, LaTeX
- **Quant & ML:** NumPy, Pandas, Scikit-Learn, XGBoost, Time-Series Analysis, Backtesting, Portfolio Construction
- **Tools:** Git/GitHub, Jupyter, Kite API, WorldQuant BRAIN, Quantiacs

PROJECTS

Systematic Trading Research *2025 – 2026*
Self Project

- Built a **modular backtester** for NIFTY500 daily and intraday data with transaction costs, slippage, turnover and drawdown analysis
- Implemented **walk-forward evaluation**, robustness testing and automated performance reporting
- Researched **cross-sectional and residual alpha signals** while

OCRFlow *2026*
Self Project

- Built a **production-oriented OCR pipeline** with modular preprocessing, OCR inference and post-processing components
- Developed **56 tests** with **94.4% coverage**, with Ruff linting, MyPy type checking and Docker Compose validation
- Designed reproducible development and deployment workflows

TE Process Fault Classification *Aug'25 – Nov'25*
CHE-657 Course Project

- Designed a **multivariate time-series fault classification pipeline** on Tennessee Eastman Process sensor data using a simulation-block representation of 600 time steps × 34 variables
- Implemented a **1D-CNN** to capture temporal dependencies; applied an 80/20 stratified split, 5-fold cross-validation and strict data-leakage prevention with standard scaling
- Achieved **78.07% mean test accuracy** across 29 fault classes, demonstrating robust generalization on unseen industrial

RNA–Protein Binding Prediction *Jan'26 – Apr'26*
CHE-629 Course Project

- Reproduced the **iDeepS dual-branch architecture** using sequence CNN, structure CNN and BiLSTM across 3 real CLIP datasets and 1 synthetic sanity dataset
- Developed **cross-attention sequence–structure fusion** and a 4-expert MoE-SwiGLU model; achieved **90.58% mean accuracy**, 0.754 F1 and 0.934 ROC-AUC
- Validated structure sensitivity on synthetic data, with structure-aware models reaching **AUC 1.0** versus 0.77 for

Interactive Portrait Shadow Removal *Jan'26 – Apr'26*
EE-655 Course Project

- Engineered a **PyTorch shadow-removal pipeline** using a frozen InceptionResNetV1 encoder and U-Net decoder with user-drawn segmentation masks
- Generated synthetic training data from **CelebA-HQ and FFHQ** and evaluated restoration using PSNR and SSIM
- Deployed an interactive **Streamlit application** for real-time

POSITIONS OF RESPONSIBILITY

Secretary, Outreach Cell, IIT Kanpur *May'24 – Apr'25*

- Led **alumni engagement, communication drives, and event coordination**; facilitated cross-generational networking and maintained institutional memory through structured outreach
- Created **standard communication templates**, improved response rates, and analyzed alumni engagement metrics to strengthen outreach effectiveness

RELEVANT COURSES

(* ONLINE COURSE, ** ONGOING)

BAYESIAN METHOD IN FINANCE
COMPUTER METHODS FOR ENGINEERS
COMPUTER VISION AND DEEP LEARNING
LINEAR ALGEBRA
FUNDAMENTALS OF COMPUTING - I
ARTIFICIAL INTELLIGENCE IN SYSTEMS BIOLOGY
ORDINARY DIFFERENTIAL EQUATIONS
Data Analytics with Python*
MACHINE LEARNING IN CHEMICAL ENGINEERING